

Follow-along guide for Wazuh setup

This reading will introduce you to **Wazuh**, a powerful, open source platform that serves as a modern Security Information and Event Management (SIEM) tool. Like other SIEM solutions, Wazuh is designed to collect, aggregate, and analyze massive amounts of security data to help identify and respond to threats.

For these reasons, Wazuh is an ideal alternative to other well known SIEM tools like **Splunk** and **Google SecOps**. While those platforms are industry leaders, their free trials and student access have become increasingly restricted, often requiring a business or organizational email address.

Fortunately, Wazuh provides a perfect solution. As a fully open source and free platform, it allows anyone to gain valuable, hands on experience with the same core skills used by professional security analysts, without facing any financial or access barriers. You can perform all the essential tasks of a security professional, including log analysis, threat detection, and incident investigation, on a system that you control completely.

Note: The following reading is an optional supplement to the course item, [Activity: Perform a query with Wazuh](#). If you choose not to read the material, please download the [tutorialdata.zip](#) file. Both this reading and the following activity are optional and will not affect your completion of the course. You may choose to skip this reading and/or the activity for any reason, and continue progressing through the remainder of the course.

Link to supporting materials: [tutorialdata.zip](#) file.

This reading includes detailed instructions for getting started with the following course item,

[Activity: Perform a query with Wazuh](#). Use this reading for step-by-step instructions on how to set up a Wazuh virtual machine, configure it, and ingest data for analysis.

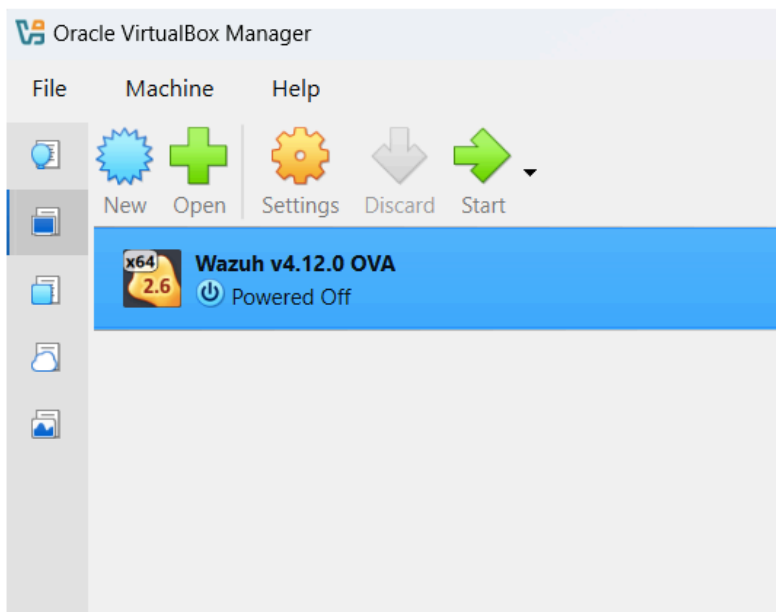
This guide provides a proven method to successfully install and use Wazuh, including key troubleshooting steps that other learners have found helpful.

Instructions

Part 1 - Deploy the Wazuh Virtual Machine

To use Wazuh, you will set up a local server on your computer using a virtual machine. This does not require a company email or any paid service.

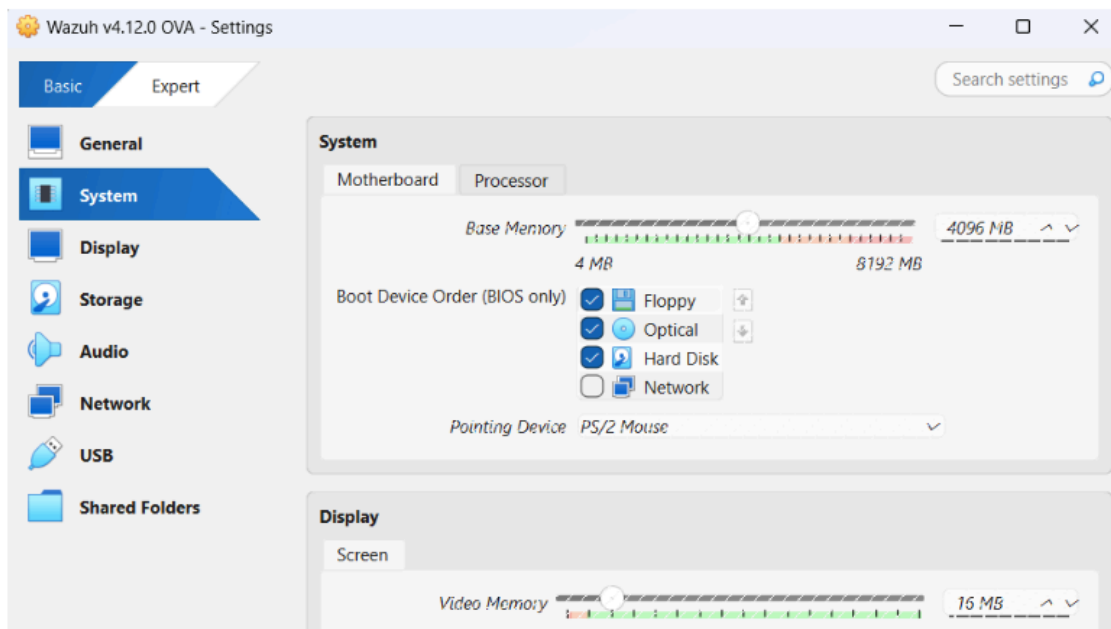
1. Download and install [Oracle VirtualBox](#) to run the virtual machine.
2. Search online for "**Wazuh virtual machine OVA**" to find and download the pre-built Wazuh server file.
3. Open VirtualBox and go to **File > Import Appliance...**
4. Select the Wazuh .ova file you downloaded. Follow the prompts to import it.

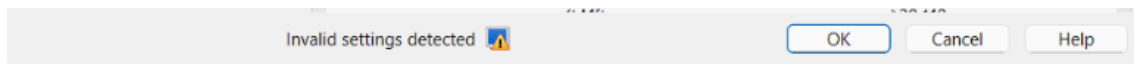


Part 2 - Configure the Virtual Machine

Before you start the VM, you must configure its memory to prevent errors.

1. In the main VirtualBox window, select your imported Wazuh VM.
2. Click the **Settings** button.
3. Go to the **System** tab. Adjust the **Base Memory** slider to **4096 MB (4 GB)**. This is a critical step to ensure your computer has enough resources to run both your operating system and the VM.
4. Click **OK** to save the setting.

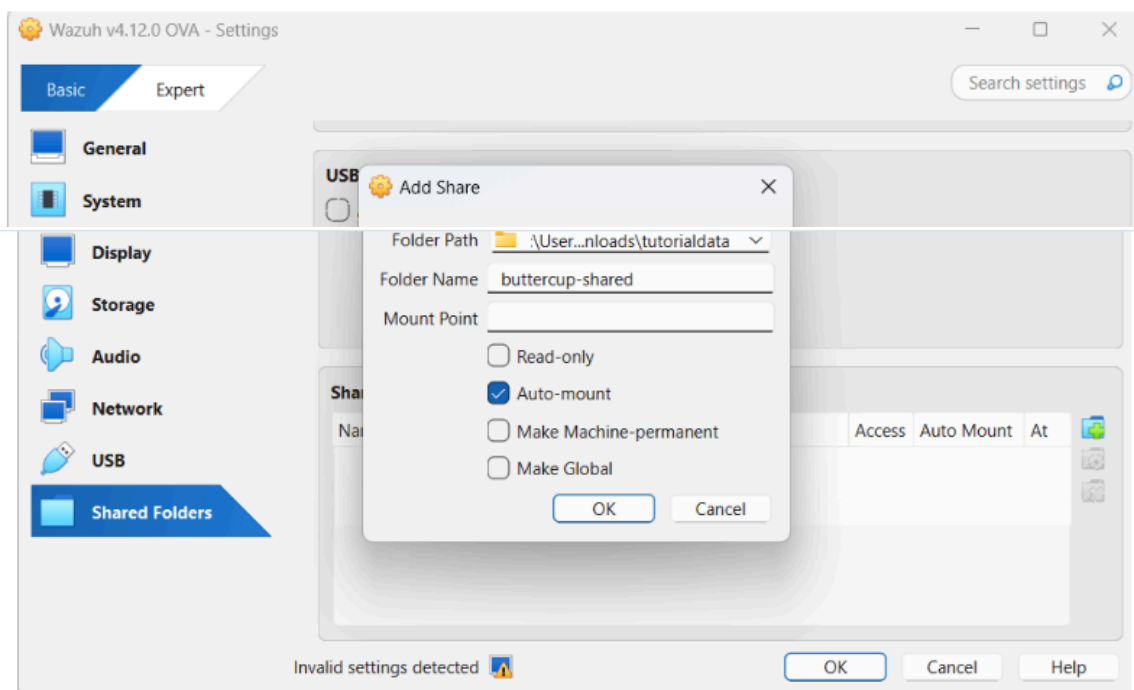




Part 3 - Access the Wazuh Dashboard

Now you will start the VM, configure a shared folder to get your data inside, and then access the dashboard from your browser.

1. **Start the VM:** Click the **Start** button in VirtualBox. Once it boots, you can press the **right Ctrl key** to get your mouse back from the VM's window.
2. **Set Up Shared Folder:** From the VirtualBox menu bar, go to **Devices > Shared Folders > Shared Folders Settings...**
 - Click the **Add new shared folder** icon (green plus sign).
 - For **Folder Path**, navigate to and select the [tutorialdata](#) folder you **unzipped**.
 - For **Folder Name**, enter **buttercup-shared**. Ensure the **Auto-mount** box is checked and click **OK**.



3. **Fix Permissions:** After the VM boots, log in with **root / wazuh** (use these root credentials to access logs). Run the following commands to get the correct permissions for the shared folder:

root permissions

```
1 usermod -aG vboxsf root
2 reboot
```

4. **Log in and Access:** After the VM reboots, log back in (with root credentials above). Run the command **ip a** to find your VM's IP address. The address will likely start with 192. Open a web browser on your computer and go to https://<your_VM_IP_address>. Allow and log in to the Wazuh dashboard with admin / admin.

Part 4 - Ingest and Analyze Data

Now that your VM is running, you can get the data in for analysis.

1. **Access the Data:** In the VM's command line, run **cd /media/sf_buttercup-shared**. The sf_ prefix is added by VirtualBox to denote the shared folder.
2. **Create Ingestion File:** Run **nano ingest.yml** and copy/paste the following content. **Note:** This is a YAML file; be precise with your spacing.

ingest.yml

```
1  filebeat.inputs:
2    - type: log
3      enabled: true
4      paths:
5        - /media/sf_buttercup-shared/maillsv/*.log
6        - /media/sf_buttercup-shared/vendor_sales/*.csv
7        - /media/sf_buttercup-shared/www1/*.log
8        - /media/sf_buttercup-shared/www2/*.log
9        - /media/sf_buttercup-shared/www3/*.log
10 output.logstash:
11   hosts: ["localhost:5044"]
```



The screenshot shows the GNU nano 8.3 text editor with the file ingest.yml open. The editor's status bar at the bottom indicates it has read 11 lines. The content of the file is as follows:

```
GNU nano 8.3 ingest.yml
filebeat.inputs:
- type: log
  enabled: true
  paths:
    - /media/sf_buttercup-shared/maillsv/*.log
    - /media/sf_buttercup-shared/vendor_sales/*.csv
    - /media/sf_buttercup-shared/www1/*.log
    - /media/sf_buttercup-shared/www2/*.log
    - /media/sf_buttercup-shared/www3/*.log
output.logstash:
  hosts: ["localhost:5044"]
```

The bottom of the screen shows the nano editor's command palette with various shortcuts like ^G Help, ^O Write Out, ^F Where Is, ^K Cut, ^T Execute, ^C Location, ^X Exit, ^R Read File, ^_ Replace, ^U Paste, ^J Justify, and ^_ Go To Line.

Press **Ctrl+X**, then type **Y**, and press **Enter** to save.

3. **Run Ingestion:** Run **/usr/share/filebeat/bin/filebeat -c ingest.yml -e**. The command will process your logs and

send them to the dashboard.

4. **Verify & Analyze:** After the command finishes, go to your browser. If you don't see logs immediately, wait a few minutes and refresh.

- Go back to the Dashboard and locate the **Discover** option under the **Explore** option.
- In the time range, choose **Absolute**, select a very old start date (e.g., January 1, 2000), and click **Update**.
- In the search bar, type * and press **Enter**. You are now ready to answer the questions in the activity. **Note:** If you get less than 100 hits, you should revise the steps above.

You're done! Once your Wazuh environment is set up, you can begin the next course item, Activity: Perform a query with Wazuh.